

Decent Coaching Classes

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Worksheet

1. Write the coefficient of x^2 and the constant term of each expression. Also state whether it is a monomial, binomial or trinomial.

(a) $3x^2 - 5x + 7$

(b) $4x^2 + 9$

(c) $9x^2$

(d) $5 - 3x^2$

2. Subtract:

(a) $5x^2 - 3x + 2$ from $8x^2 + x - 6$

(b) $2ab - 3a + 4$ from $5a + ab - 7$

3. Find the following products:

(a) $3a^2(2a + 3b)$

(b) $-4xy(x - 2y + 3)$

(c) $(2a + 3b)(4a + 5b)$

(d) $(2x - 3)(x^2 + 2x - 5)$

4. Simplify and verify wherever asked:

(a) $5x(x - 2) + x(3 - 4x) + 2x^2$

(b) $(2x + 1)(x - 3) - (x + 2)(2x - 1)$

(c) Multiply $(a + 2b)(a^2 - ab + b^2)$ and verify the result for $a = 1$ and $b = -2$.

5. Answer the following:

(a) Find the HCF of $12x^2y$ and $18xy^2$.

(b) Factorise: $6x^2 - 9x$

(c) Factorise: $15a^2b - 20ab^2$

(d) Factorise: $x^2 + 6x + 9$

(e) Factorise: $4a^2 + 12ab + 9b^2$

(f) Factorise: $9p^2 + 24pq + 16q^2$

Answer Key

Q1.

- (a) Coefficient of $x^2 = 3$; constant term = 7; trinomial
- (b) Coefficient of $x^2 = 4$; constant term = 9; binomial
- (c) Coefficient of $x^2 = 9$; constant term = 0 (no constant term); monomial
- (d) Coefficient of $x^2 = -3$; constant term = 5; binomial

Q2.

- (a) $(8x^2 + x - 6) - (5x^2 - 3x + 2) = 8x^2 + x - 6 - 5x^2 + 3x - 2 = 3x^2 + 4x - 8$
- (b) $(5a + ab - 7) - (2ab - 3a + 4) = 5a + ab - 7 - 2ab + 3a - 4 = 8a - ab - 11$

Q3.

- (a) $6a^3 + 9a^2b$
- (b) $-4x^2y + 8xy^2 - 12xy$
- (c) $8a^2 + 10ab + 12ab + 15b^2 = 8a^2 + 22ab + 15b^2$
- (d) $2x^3 + 4x^2 - 10x - 3x^2 - 6x + 15 = 2x^3 + x^2 - 16x + 15$

Q4.

- (a) $5x^2 - 10x + 3x - 4x^2 + 2x^2 = 3x^2 - 7x$
- (b) $(2x^2 - 5x - 3) - (2x^2 + 3x - 2) = -8x - 1$
- (c) $(a + 2b)(a^2 - ab + b^2) = a^3 + a^2b - ab^2 + 2b^3$
 $\text{LHS} = (1 - 4)(1 + 2 + 4) = (-3)(7) = -21$
 $\text{RHS} = 1 + (1)(-2) - (1)(4) + 2(-8) = 1 - 2 - 4 - 16 = -21$
 $\text{LHS} = \text{RHS}$, hence verified.

Q5.

- (a) $\text{HCF} = 6xy$
- (b) $3x(2x - 3)$
- (c) $5ab(3a - 4b)$
- (d) $(x + 3)^2$ or $(x + 3)(x + 3)$
- (e) $(2a + 3b)^2$ or $(2a + 3b)(2a + 3b)$
- (f) $(3p + 4q)^2$ or $(3p + 4q)(3p + 4q)$